RetailPulse – Team Workflow & Project Documentation

Project Title

RetailPulse – AI-Powered Customer Analytics & Demand Forecasting Platform

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1. Project Overview

Objective

RetailPulse is an end-to-end Data Science and Analytics project developed for retail businesses to help them make data-driven decisions using Machine Learning and Analytics.

The platform analyzes customer behavior, predicts future sales demand, identifies customers likely to churn, and provides inventory optimization recommendations.

The final output will be an interactive Streamlit dashboard containing multiple analytics pages along with deployed machine learning models.

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2. Business Problem

Retail businesses face several major challenges:

- Inaccurate sales forecasting
- Overstocking and understocking
- Poor customer retention
- Lack of customer insights
- Revenue loss due to inefficient inventory management

RetailPulse solves these issues using Data Science, Machine Learning, and Analytics.

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3. Main Modules of the Project

The project is divided into four major modules:

Module 1 – Sales Analytics

Provides insights about:

- Revenue trends
- Monthly sales
- Best-selling products
- Country-wise performance
- Product performance

Module 2 – Customer Analytics

Provides:

- Customer segmentation
- RFM analysis
- Customer behavior analysis
- Churn prediction
- High-value customer identification

Module 3 – Demand Forecasting

Predicts:

- Future sales demand
- Product demand trends
- Seasonal sales patterns
- Forecast graphs for business planning

Module 4 – Inventory Optimization

Provides:

- Stock recommendations
- Reorder quantity suggestions
- Low inventory alerts
- Overstock analysis

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4. Dashboard Pages

The Streamlit dashboard will contain 4 pages:

Page 1 – Sales Dashboard

Features:

• Total Revenue KPI
• Monthly Sales Trend
• Top Products
• Country-wise Revenue
• Daily/Monthly Sales Analysis

Page 2 – Customer Dashboard

Features:

• RFM Analysis
• Customer Segments
• Churn Risk Analysis
• Cluster Visualization
• Customer Spending Insights

Page 3 – Forecast Dashboard

Features:

• Sales Forecast Graph
• Future Demand Prediction
• Seasonal Trend Analysis
• Forecast Comparison

Page 4 – Inventory Dashboard

Features:

• Inventory Status
• Suggested Reorder Quantity
• Overstock Alerts
• Understock Alerts
• Product Demand Recommendations

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5. Datasets Used

Primary Datasets

1. Online Retail Dataset (UCI)

2. Online Retail II Dataset (2 Years Data)
3. Kaggle Online Retail Dataset

Important Dataset Columns

- InvoiceNo
- StockCode
- Description
- Quantity
- InvoiceDate
- UnitPrice
- CustomerID
- Country

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6. Technology Stack

Category	Technologies
Programming Language	Python
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn, Plotly
Machine Learning	Scikit-learn, XGBoost
Forecasting	Prophet, ARIMA, LSTM
Dashboard	Streamlit
Deployment	Streamlit Cloud / Render
Version Control	GitHub
Notebook Environment	Jupyter Notebook

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7. Complete Workflow of the Project

Step 1 – Dataset Collection

Objective

Collect retail sales datasets for customer analytics and forecasting.

Tasks

- Download datasets from UCI and Kaggle
- Understand dataset structure
- Identify important columns
- Verify dataset quality

Output

- Raw dataset files

Step 2 – Import Required Libraries

Objective

Set up the Python environment with required libraries.

Libraries Used

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.cluster import KMeans
from prophet import Prophet
import streamlit as st
```

Output

- Working Python environment

Step 3 – Load Dataset

Objective

Read the dataset into Python.

Tasks

- Load CSV/Excel files
- Check dataset dimensions
- Understand datatypes
- Display sample records

Example

```
pd.read_csv('online_retail.csv')
```

Output

- Loaded dataframe

Step 4 – Data Cleaning

Objective

Prepare high-quality data for analysis.

Tasks

- Remove missing values
- Remove duplicate records
- Handle cancelled orders
- Convert date columns
- Remove invalid quantities
- Handle outliers

Important Cleaning Operations

- Drop null CustomerID
- Remove negative quantities
- Convert InvoiceDate to datetime format

Output

- Cleaned dataset

Step 5 – Feature Engineering

Objective

Create meaningful features for machine learning models.

Features Created

Total Price

```
TotalPrice = Quantity * UnitPrice
```

RFM Features

Recency

Days since last purchase

Frequency

Total number of purchases

Monetary

Total amount spent by customer

Additional Features

- Monthly sales
- Customer lifetime value
- Rolling averages
- Purchase frequency

Output

- Feature engineered dataset

Step 6 – Exploratory Data Analysis (EDA)

Objective

Understand patterns, trends, and customer behavior.

Analysis Performed

Sales Analysis

- Monthly sales trends
- Revenue analysis
- Top-selling products

- Seasonal trends

Customer Analysis

- Customer purchase frequency
- Country-wise customers
- High-value customers

Visualization Types

- Bar Charts
- Pie Charts
- Heatmaps
- Line Graphs
- Correlation Matrix

Output

- EDA notebook
- Business insights
- Visual reports

Step 7 – Customer Segmentation (RFM Analysis)

Objective

Group customers based on purchasing behavior.

Process

RFM Calculation

Recency

How recently a customer purchased

Frequency

How frequently customer purchases

Monetary

How much customer spends

Clustering

Apply:

- KMeans Clustering
- DBSCAN (optional)

Customer Groups

- Premium Customers
- Loyal Customers
- Regular Customers
- At-risk Customers
- Low-value Customers

Output

- Segmented customers
- Cluster visualizations

Step 8 – Demand Forecasting

Objective

Predict future product demand and sales.

Models Used

- Prophet
- ARIMA
- LSTM (optional advanced model)

Forecasting Tasks

- Daily sales forecasting
- Monthly sales prediction
- Trend analysis
- Seasonality analysis

Forecast Metrics

- MAE
- RMSE
- MAPE

Output

- Forecast graphs
- Predicted demand values

Step 9 – Churn Prediction

Objective

Identify customers likely to stop purchasing.

Features Used

- Recency
- Frequency
- Monetary
- Purchase intervals
- Customer activity

Models Used

- XGBoost
- Random Forest
- Logistic Regression

Prediction Classes

- High churn risk
- Medium churn risk
- Low churn risk

Evaluation Metrics

- Accuracy
- Precision
- Recall
- AUC Score

Output

- Churn prediction model
- Customer risk analysis

Step 10 – Inventory Optimization

Objective

Recommend stock levels based on forecasted demand.

Logic

```
Suggested Stock = Forecasted Demand - Current Inventory
```

Features

- Reorder quantity
- Low stock alerts
- Overstock alerts
- Demand-based recommendations

Output

- Inventory recommendations

Step 11 – Streamlit Dashboard Development

Objective

Create an interactive business analytics dashboard.

Dashboard Features

Sales Dashboard

- Revenue KPIs
- Trend graphs
- Product analysis

Customer Dashboard

- Customer segmentation
- Churn analysis
- Customer insights

Forecast Dashboard

- Future demand prediction

- Forecast visualization

Inventory Dashboard

- Inventory recommendations
- Stock alerts

Output

- Fully functional dashboard

Step 12 – Deployment

Objective

Deploy the application online.

Deployment Platforms

- Streamlit Cloud
- Render
- AWS (optional)

Deployment Tasks

- Upload code to GitHub
- Configure requirements.txt
- Deploy Streamlit application
- Test live application

Output

- Live deployed application

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8. Team Structure and Work Division

Since the team contains 4 members from the Data Science & Analytics domain, work should be divided module-wise.

Member 1 – Data Processing & EDA Lead

Responsibilities

- Dataset collection
- Data cleaning
- Feature engineering
- Exploratory Data Analysis
- Visualization preparation

Deliverables

- Clean dataset
- EDA notebook
- Business insights report

Technologies Used

- Pandas
- NumPy
- Matplotlib
- Seaborn

Member 2 – Customer Analytics Lead

Responsibilities

- RFM Analysis
- Customer segmentation
- Churn prediction
- Customer behavior analysis

Deliverables

- Segmentation model
- Churn prediction model
- Cluster analysis

Technologies Used

- Scikit-learn
- XGBoost
- KMeans

Member 3 – Forecasting & Inventory Lead

Responsibilities

- Time-series analysis
- Demand forecasting
- Forecast visualization
- Inventory optimization logic

Deliverables

- Forecasting model
- Inventory recommendation system

Technologies Used

- Prophet
- ARIMA
- LSTM

Member 4 – Dashboard & Deployment Lead

Responsibilities

- Streamlit dashboard
- UI integration
- Model integration
- Deployment
- GitHub management

Deliverables

- Working dashboard
- Live deployment link
- Demo presentation

Technologies Used

- Streamlit
- GitHub
- Streamlit Cloud

9. Weekly Execution Plan

Week 1 – Data Preparation & EDA

Goals

- Collect datasets
- Clean data
- Perform EDA
- Create initial visualizations

Tasks

- Data cleaning
- Missing value handling
- Feature engineering
- Trend analysis

Week 2 – Machine Learning Development

Goals

- Build segmentation model
- Build forecasting model
- Build churn model

Tasks

- RFM analysis
- Clustering
- Time-series forecasting
- Churn prediction

Week 3 – Dashboard Integration

Goals

- Create dashboard pages
- Integrate all models
- Add visualizations

Tasks

- Streamlit UI development

- Graph integration
- Testing dashboard functionality

Week 4 – Deployment & Finalization

Goals

- Deploy project
- Final testing
- Documentation
- PPT and demo preparation

Tasks

- GitHub cleanup
- Deployment
- Demo recording
- Final report creation

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10. GitHub Workflow

Repository Structure

```
RetailPulse/
│
├── data/
├── notebooks/
├── src/
├── dashboard/
├── models/
├── reports/
├── requirements.txt
└── README.md
```

Recommended Branch Structure

```
main
feature/eda
feature/churn-model
```

```
feature/forecasting
feature/dashboard
```

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11. Deliverables

The team must submit:

Mandatory Deliverables

1. Project Report PDF
2. GitHub Repository
3. Live Demo Link
4. Streamlit Dashboard
5. Demo Video
6. README Documentation

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12. Key Performance Metrics

Module	Metrics
Forecasting	MAPE, RMSE
Churn Prediction	Accuracy, AUC
Segmentation	Silhouette Score
Dashboard	User Experience

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13. Challenges Expected

Technical Challenges

- Handling missing values
- Time-series forecasting complexity
- Customer churn labeling
- Dashboard integration

Team Challenges

- Merge conflicts

- Coordination issues
- Model integration
- Deployment debugging

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14. Best Practices

Technical Best Practices

- Use modular code
- Maintain clean notebooks
- Use GitHub regularly
- Keep reusable functions
- Save trained models

Team Best Practices

- Daily updates
- Weekly review meetings
- Shared documentation
- Common coding standards

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15. Final Expected Outcome

At the end of the internship, the team will have:

- A complete industry-level analytics project
- Multiple machine learning models
- An interactive business dashboard
- A deployed web application
- Practical experience in end-to-end Data Science workflow
- Strong portfolio project for internships and placements

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16. Conclusion

RetailPulse is a complete end-to-end Data Science and Analytics solution designed to solve real-world retail business problems.

The project combines:

- Data Analytics

- Machine Learning
- Forecasting
- Customer Intelligence
- Dashboard Development
- Deployment

This project provides hands-on industry-level experience in:

- Data preprocessing
- Machine learning
- Time-series forecasting
- Customer analytics
- Business intelligence
- Dashboard deployment

The successful completion of this project will demonstrate strong practical skills in Data Science and Analytics.